Ch 5.1.4-5: More Cross-Validation

Lecture 12 - CMSE 381

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Fri, Oct 7, 2022

Announcements

Last time:

- k-fold CV

This lecture:

- More k-fold CV
- Bias-Variance Tradeoff
- CV for classification

Announcements:

- Homework #4 is posted, Due Monday
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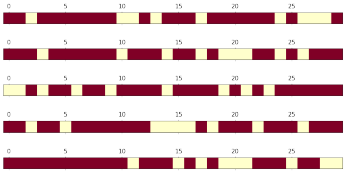
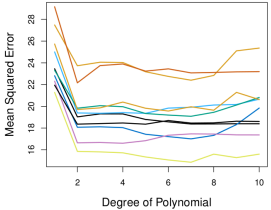
Section 1

k-fold CV

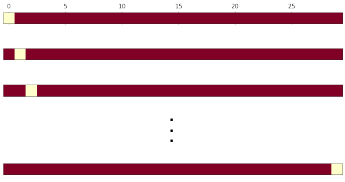
Approximations of Test Error

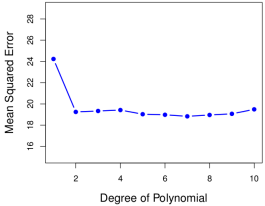
Validation Set



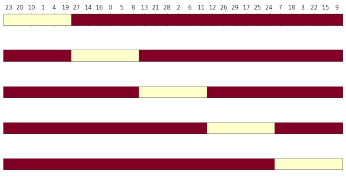


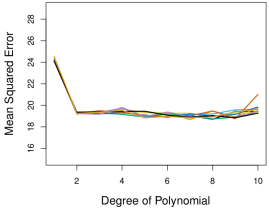
LOOCV





K-fold CV

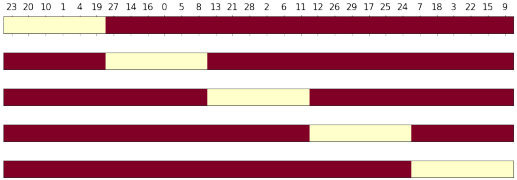




Definition of k-fold CV

shuffle data then break up into k groups

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points



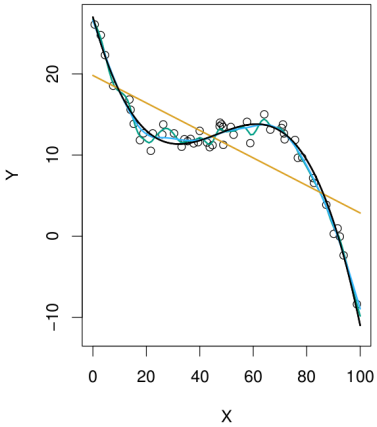
- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{MSE}_i = \frac{1}{\ell}\sum_{(x_j,y_j)\in U_i}(y_j - \hat{y}_j)^2$
- Rinse and repeat

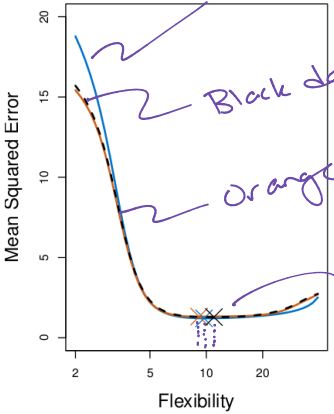
Return

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k}\mathrm{MSE}_i$$

If $k=n \Rightarrow$ LOOCV

Comparison with simulated data: Ex 3





Blue: true test MSE

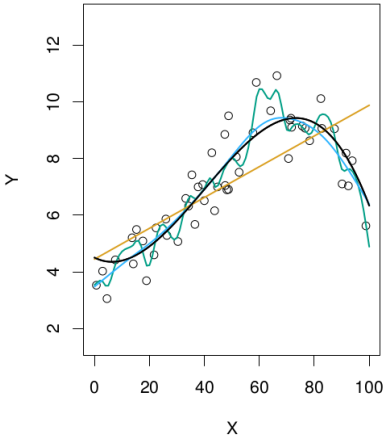
Black dashed: LOOCV

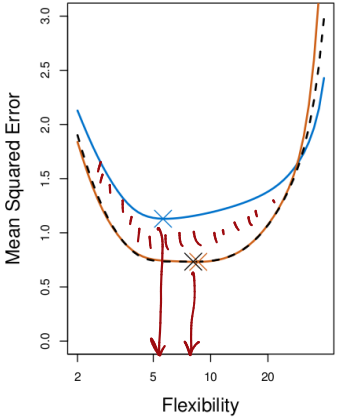
Orange: 10-fold CV

X = location of minimum of test error approx.

All minimums occur at approx the same flexibility value.

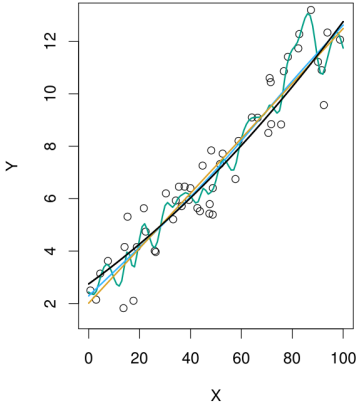
Comparison with simulated data: Ex 1

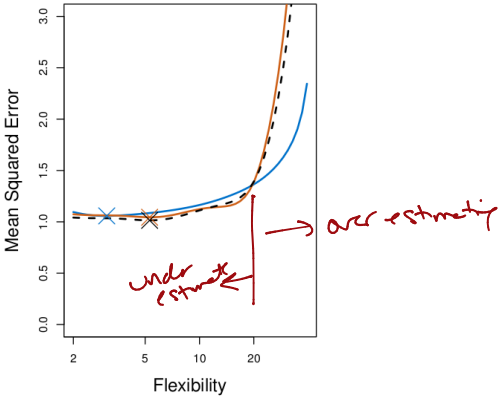




* LOOCV and 10-fold CV underestimates of the true value

Comparison with simulated data: Ex 2





Takeaways from the examples

- All examples: LOOCV and 10-fold CV were very similar
- Often only care about the location of the minimum rather than the value of minimum, and these are similar in all cases.

x

y

Bias-Variance Tradeoff: Bias

$Y = f(X) + \varepsilon$

$$\overbrace{E(y_0 - \hat{f}(x_0))^2}^{\text{true test MSE error}} = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

Irreducible error

Validation set overestimates the test error because it uses small subset of data

↑ small k

K-fold CV medium level of bias b/c training set has $\frac{k-1}{k}(n)$ data pts....

↓ larger K

LOOCV gives approximately unbiased estimate since use almost all data every time

Bias-Variance Tradeoff: Variance

Empirically

$$E(y_0 - \hat{f}(x_0))^2 = \mathrm{Var}(\hat{f}(x_0)) + \left[\mathrm{Bias}(\hat{f}(x_0))\right]^2 + \mathrm{Var}(\varepsilon)$$

$k = 5$ or 10

LOOCV has higher variance than k-fold

- avg the outputs of n fitted models
- done on almost identical observations
- high correlation with each other → higher variance

k-fold

- k fitted models, somewhat less correlated with each other

Coding - Build a plot for train/test scores vs flexibility

~~Fix degree d~~

for degree in d = 1, —, 6

for whatever in k fold

train test split

train model

test model

Get average of MSES

1

Coding - Build a plot for train/test scores vs flexibility

polynomial matrix $X = \begin{bmatrix} | & | & | & \\ x & x^2 & x^3 & \dots \\ | & | & | & \end{bmatrix}$

for $d = 1, \dots, 6$:

$X = \begin{bmatrix} | & | & | & & | \\ x & x^2 & x^3 & \dots & x^d \\ | & | & | & & | \end{bmatrix}$

for train/split in kfold:

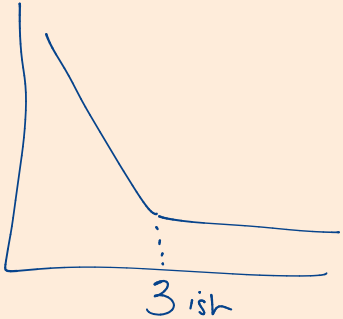
$\dashrightarrow$ X[train], X[test]

Y[train] y[test]

LinearRegression().fit(X_train, y_train)

mse = lin reg.score(X_test, y_test)

avg mse into some list



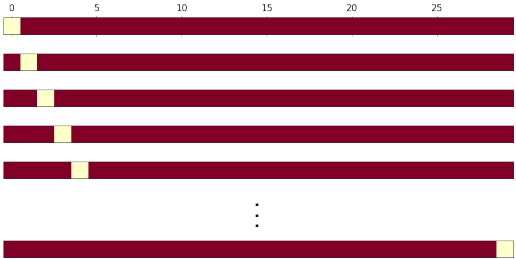
Section 2

CV for Classification

Setup: LOOCV

- Remove ith point (x_i, y_i) and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{Err}_i = \mathrm{I}(y_j \neq \hat{y}_j)$

- Rinse and repeat



Return

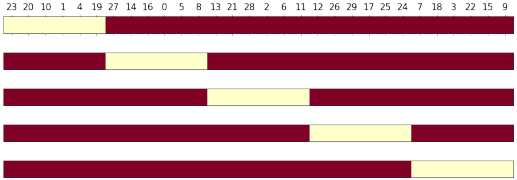
$$CV_{(n)} = \frac{1}{n}\sum_{i=1}^{n} \mathrm{Err}_i$$

Setup: k-fold

- Randomly split data into k-groups (folds)
- Approximately equal sized. For the sake of notation, say each set has ℓ points

- Remove ith fold U_i and reserve for testing.
- Train the model on remaining points
- Calculate $\mathrm{Err}_i = \frac{1}{\ell} \sum_{(x_j, y_j) \in U_i} \mathrm{I}(y_j \neq \hat{y}_j)$

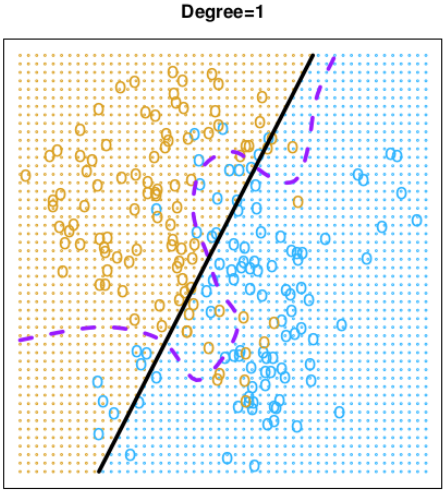
- Rinse and repeat



Return

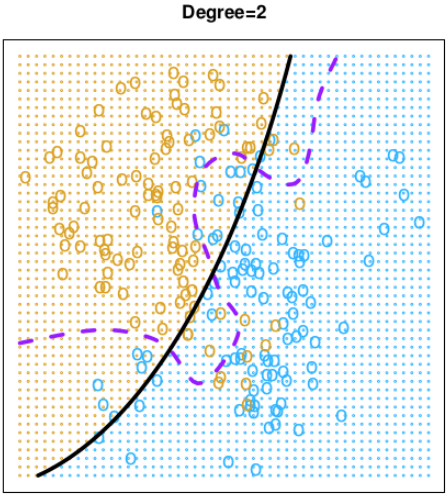
$$CV_{(k)} = \frac{1}{k} \sum_{i=1}^{k} \mathrm{Err}_i$$

Example on simulated data: Linear



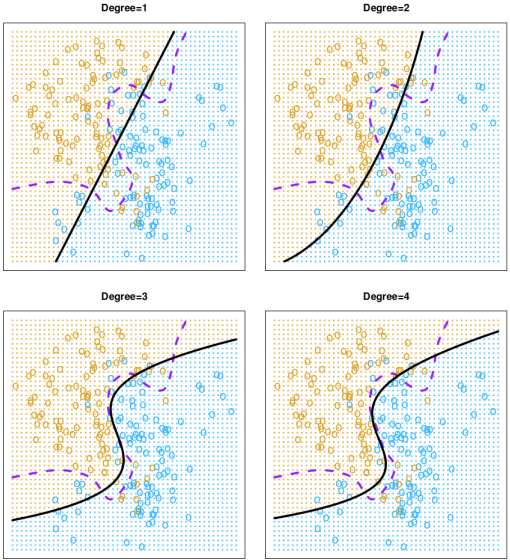
- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - $\log(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_2$
  - Error rate: 0.201

Example on simulated data: Quadratic logistic regression



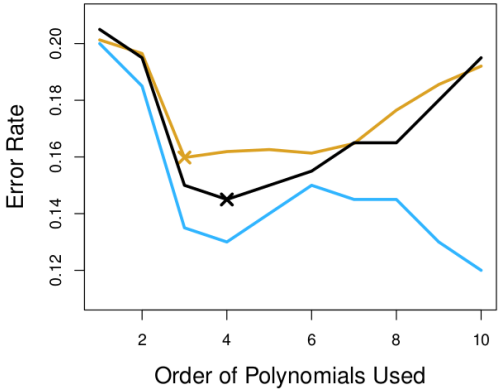
- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - $\log(p/(1-p)) = \beta_0 + \beta_1 X_1 + \beta_2 X_1^2 + \beta_3 X_2 + \beta_4 X_2^2$
  - Error rate: 0.197

Example on simulated data: all the polynomials!



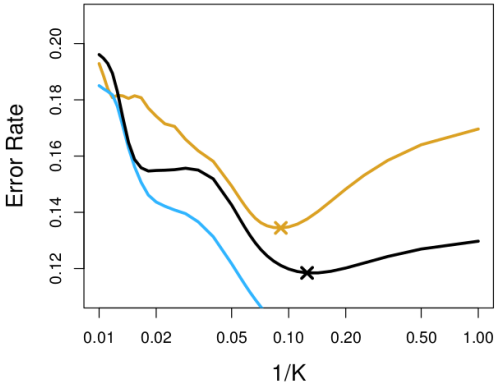
- Purple: Bayes decision boundary.
  - Error rate: 0.133
- Black: Logistic regression
  - Deg 1 Error rate: 0.201
  - Deg 2 Error rate: 0.197
  - Deg 3 Error rate: 0.160
  - Deg 4 Error rate: 0.162

Decide degree based on CV



- Test error (brown)
- Training error (blue)
- 10-fold CV error (black)

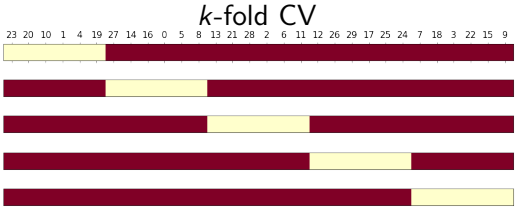
Similar game for KNN



- Test error (brown)
- Training error (blue)
- 10-fold CV error (black)

Coding - k-fold for Classification section

TL;DR



$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{MSE}_i$$

Use $k = 5$ or 10 usually

k-fold CV for classification

$$\mathrm{Err}_i = \mathrm{I}(y_j \neq \hat{y}_j)$$

$$CV_{(k)} = \frac{1}{k}\sum_{i=1}^{k} \mathrm{Err}_i$$

Next time

10	M	Oct 3	Leave one out CV	5.1.1, 5.1.2	
11	W	Oct 5	k-fold CV	5.1.3	
12	F	Oct 7	More k-fold CV, k-fold CV for classification	5.1.4-5	
13	M	Oct 10	Resampling methods: Bootstrap	5.2	HW #4 Due
14	W	Oct 12			
15	F	Oct 14	Subset selection	6.1	
16	M	Oct 17	Shrinkage: Ridge	6.2.1	HW #5 Due
17	W	Oct 19	Shrinkage: Lasso	6.2.2	
18	F	Oct 21	Dimension Reduction	6.3	
	M	Oct 24	No class - Fall break		
19	W	Oct 26	More dimension reduction; High dimensions	6.4	
20	F	Oct 28	Polynomial & Step Functions.	7.1,7.2	HW #6 Due
	M	Oct 31	Review		
	W	Nov 2	***Midterm #2***		